

separate models that fulfill each objective. Models aligned with human expectation should shed human tendencies such as resource-rationality, i.e. sacrificing quality to reduce effort (Evans et al., 2015; Alanqary et al., 2021; Lieder and Griffiths, 2020), while models designed to simulate humans should retain them. By recognizing the difference between people’s expectations and behavior, we provide support for developing more specific alignment objectives grounded in social science.

We hypothesize that certain training paradigms may be more suited towards aligning to human expectations, while others favor alignment with human behavior. For instance, high-quality written responses used in Supervised Fine-Tuning may teach the LLM to mimic the original human writers, aligning outputs towards human behavior. On the other hand, when humans provide preferences for RLHF in a chat setting, their judgement might reflect what the human rater expects from the LLM.

Implications for simulating humans using LLMs. There is a growing literature investigating whether we can use LLMs to simulate humans for various applications (Park et al., 2023), such as acting as mock participants in human studies (Argyle et al., 2023; Aher et al., 2022; Hämäläinen et al., 2023), collecting public opinion (Chu et al., 2023; Kim and Lee, 2023; Sun et al., 2024), and helping provide realistic reactions to assist people’s communication (Liu et al., 2023; Shaikh et al., 2024; Lin et al., 2024; Shin and Kim, 2023). As our experiments show that LLMs make decisions more rationally than people do, and also predict people’s decisions to be more rational than they are, current LLMs are fundamentally misaligned for the task of simulating human choices. Developing policy recommendations or designing experiments based on LLM-simulated choices may be misleading — similar to concerns about the use of an overly rational “homo economicus” originally raised by researchers in behavioral economics (Tversky and Kahneman, 1974; Kahneman and Tversky, 1979).

Limitations. Our experiments focused on a subset of all tasks for which we could evaluate models’ rationality. This subset was carefully selected to be representative of the classic literature, but could be expanded upon in future research. In particular, our experiments used controlled, abstract domains that do not guarantee generalization to real-world contexts. While this is the same challenge faced by all human experiments, it is uniquely challenging for work done on LLMs due to their black-box nature and sensitivity to input prompts (Sclar et al., 2023). Furthermore, like all human data, the psychological datasets that we compare LLMs against were potentially subject to sampling bias and it is unknown whether they fully represent the true distribution of human choices or inferences.

Another limitation is that we use simple simulation paradigms for testing LLMs’ capabilities. While we conduct an ablation using demographic prompts, other methods such as adding personal details or traits for more realistic variation have been proposed (Zhou et al., 2024b; Hu and Collier, 2024). The `choices13k` dataset does not contain such information about subjects, making a representative inclusion of these features impossible. However, future work could conduct a more comprehensive analysis of different simulation methods to see if they alter LLMs’ implicit decision-making models.

A third limitation to our work is that we do not directly compare LLM predictions of human decisions with human predictions of the same decisions. The psychology literature has shown that there can be differences between when people perform the risky choice task and when they predict how others would perform the same task: Faro and Rottenstreich (2006) found that people’s predictions of others’ choices are closer to risk neutrality than choices of their own; when people are risk seeking, they predict that others will be risk seeking but less so; and where people are risk averse, they predict that others will be risk averse but also less so. Collecting a dataset of people’s predictions about others’ decisions could allow us to make a quantitative comparison of humans and LLMs in this setting.

Future directions. A peculiar observation from the inverse modeling setting is that LLMs’ inferences were consistently more correlated with people than they were with rational models. This suggests that LLMs may potentially capture aspects of human behavior that are not present in existing theoretical models. While we have focused on using theories and paradigms from psychology to analyze LLMs, there may also be opportunities to use LLMs to refine existing theories about people. More generally, our results show how studies and theories originating in psychology and cognitive science can help quantify the behavior of LLMs. These fields offer many more opportunities to compare the behavior of LLMs with humans, helping us towards understanding (both) these complex yet extremely capable systems.

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